

Dynamics of Corporate Credit Markets, Employment and Wages: Evidence from Colombia

Aristizábal-Ramírez & Posso (2021, *JMP*)

Felipe Camêlo

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- A huge literature documenting the link between firms' access to credit and their real outcomes
- However, most of the work has focused on times of financial crises
- Question:
 - How does access to credit affect investment, employment and wages when neither banks or firms are facing extraordinary conditions?

This Paper

This paper: New Data + Empirical Exercise + Model

→ New Data:

- Construct a rich annual employee-employer-credit-bank database using Colombian administrative data from the period 2008-2018

→ Empirics:

1. Estimate firm-level idiosyncratic supply shocks
2. Using LPs, use these shocks to estimate impact on firms' borrowing, investment, employment and average wages
3. Using quantile regressions, evaluate heterogeneous effects across the wage distribution (not today)

→ Model

- Real Small Open Economy with working capital constraints, a liquid asset, banks, two types of labor (skilled and unskilled), and a frictional labor market

Structure of the Presentation

Data

Identifying Shocks to Credit Supply

The Effect of Credit Shocks on Investment, Employment and Wages

Unpacking The Effect of Credit Shocks: The Role of Liquidity

Model

Conclusion and Comments

Data

A combination of three different Colombian administrative data sources

- Banking Data (Superintendencia Financiera)
 - Quarterly balance data from banks and credit institutions: allows to track loans between bank b and firm f
 - Coverage: January 2008 to December 2018; 16 Banks
- Firm Data (Superintendencia de Sociedades)
 - Annual financial reports from corporations
 - Coverage: 2008 to 2015; 10835 firms
- Worker Data (PILA system)
 - Firms' monthly social security payment reports
 - Coverage: 2008 to 2018; 3321640 workers

Summary Statistics

	Mean	Std. Dev	P95	P5	N
Firms					
Employment	121	526	429	3	10,835
Leverage	0.38	5.41	0.73	0.02	10,835
Equity to Assets	3.62	86.16	8.28	1.17	10,835
Capital	16.24	222.11	42.79	0.15	10,835
Sales	10.96	141.50	32.69	0.22	10,835
Banking Shock	0.05	0.13	0.27	-0.12	10,835
Workers					
Wage	542.96	625.46	1,683.68	166.71	3,321,640
Age	34.83	10.32	54.00	21.00	3,321,640
Male	0.59	0.49	1.00	0.00	3,321,640

Note: *N* is the total number of firms or workers. Employment: Average number of workers per year.

Identifying Shocks to Credit Supply

Identifying Shocks to Credit Supply - Part 1

- An application of the framework proposed by Amiti & Weinstein (2018)
 - Credit-supply shocks as the firms' common change in borrowing from a particular bank
- Suppose a firm f borrows d_{fb} from a bank b . At each t , we can express the change in borrowing as

$$\Delta d_{fbt} = \alpha_{ft} + \beta_{bt} + \varepsilon_{fbt}$$

- To allow for the creation and destruction of bank-firm relationships, we can renormalize the equation above so that

$$\Delta d_{fbt} = c_t + \tilde{\alpha}_{ft} + \tilde{\beta}_{bt} + \varepsilon_{fbt}$$

Identifying Shocks to Credit Supply - Part 2

- Let

$$\theta_{fbt} = \frac{d_{fbt-1}}{\sum_b d_{fbt-1}}$$

then we define a credit supply shock as

$$Supply\ Shock_{ft} = \sum_b \theta_{fbt-1} \hat{\beta}_{bt}$$

i.e. credit-supply shock $\rightarrow$ a change in loan supply to firm f relative to the average change in credit supply

- Key Remarks
 - Method works only if there is sufficient overlap between banks and firms
 - Idiosyncratic changes in the credit supply, different from previous literature that focuses on aggregate shocks

The Effect of Credit Shocks on Investment, Employment and Wages

The Effect of Positive Credit-Supply Shock on Firm Observables

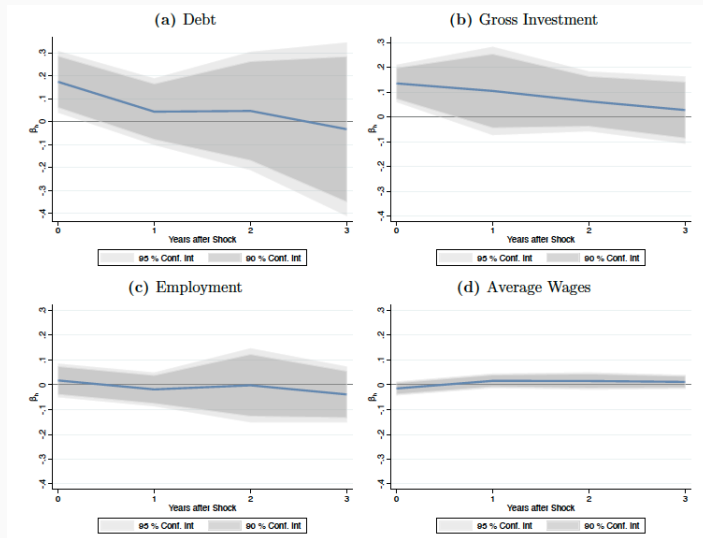
- Let Y_{fj} be an outcome of interest for firm f in period j
- At the firm level, we estimate

$$\log Y_{ft+h} - \log Y_{ft} = \beta_{0h} + \beta_h \text{Supply Shock}_{ft} + X_{ft-1} \Gamma_h + \alpha_{jth} + \alpha_{fh} + \varepsilon_{fth}$$

where β_h is the coefficient of interest

- This works as a Jordà (2005) local projection to a setup with panel data and idiosyncratic shocks

The Effect of Positive Credit-Supply Shock on Firm Observables - IRFs



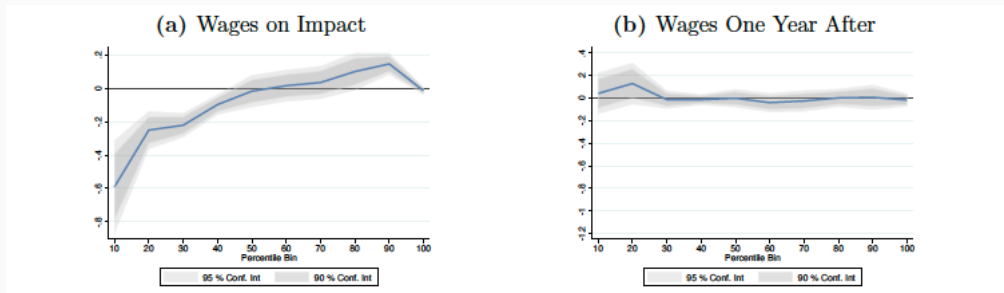
The Effect of Positive Credit-Supply Shock on Wages

- Let w_{ifj} be the wage of worker i at firm f in period j
- To estimate the effect of the credit supply shock on wage growth, we classify each worker in a wage decile $decile_{ij}$
- Then, we estimate

$$\log(w_{ift+h}) - \log(w_{ift-1}) = \beta_h \text{Supply Shock}_{ft} + \beta_{hd} \text{Supply Shock}_{ft} \times decile_{it-1} + X_{ift-1} \Gamma_h + \alpha_{fth} + \alpha_{ih} + \varepsilon_{ifth}$$

where $\beta_h + \beta_{hd}$ are the coefficients of interest

The Effect of Positive Credit-Supply Shock on Wages - Results



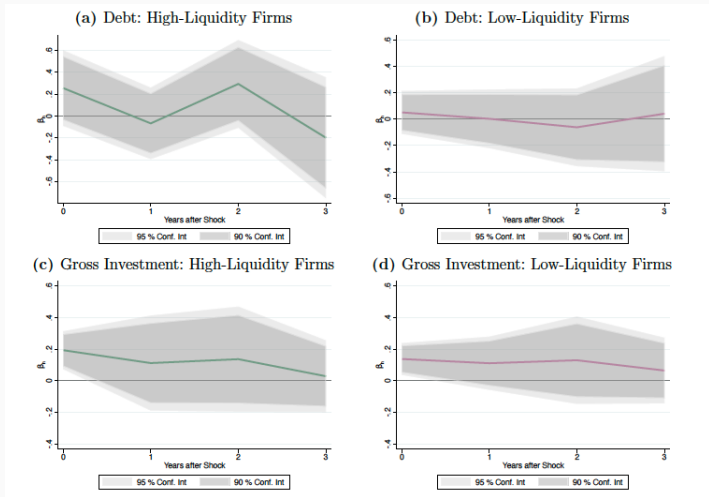
The horizontal axis shows the wage deciles, and the vertical axis shows the estimated coefficient of the total effect of a positive credit-supply shock in the wage growth rate

Unpacking The Effect of Credit Shocks: The Role of Liquidity

The Role of Liquidity

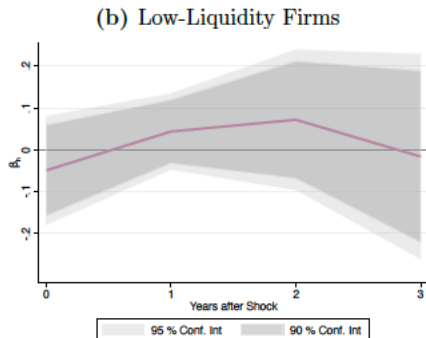
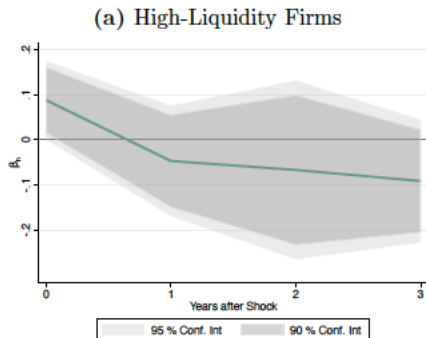
- Credit supply shock $\implies$ investment $\uparrow$, but bottom half of the wage distribution $\downarrow$
- These two simultaneous effects as a sign of a change in the demand for low skilled workers
- In the presence of working capital constraints, instead of expanding in scale, less liquid firms could choose to substitute them for capital
- In the paper, liquidity is defined as the ratio of cash and short-term investments to assets.
 - Sample broken down in two: high (above average) vs low (below average) liquidity firms

“Irrelevance” of liquidity for Debt and Investment



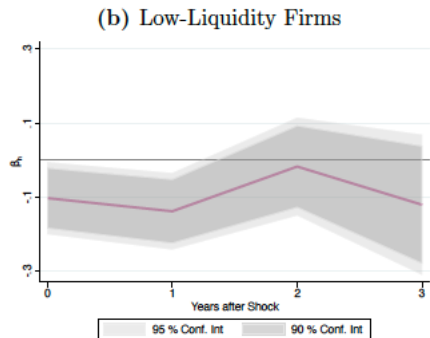
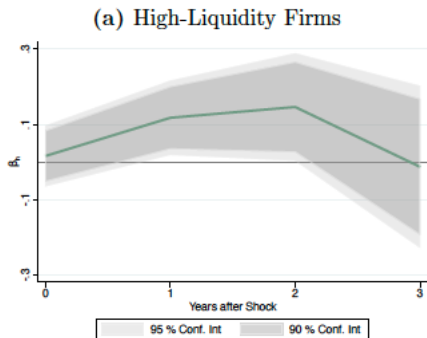
Differential Impact for Employment: Working Capital Constraints?

Working capital as the ratio of short-term assets to short-term liabilities

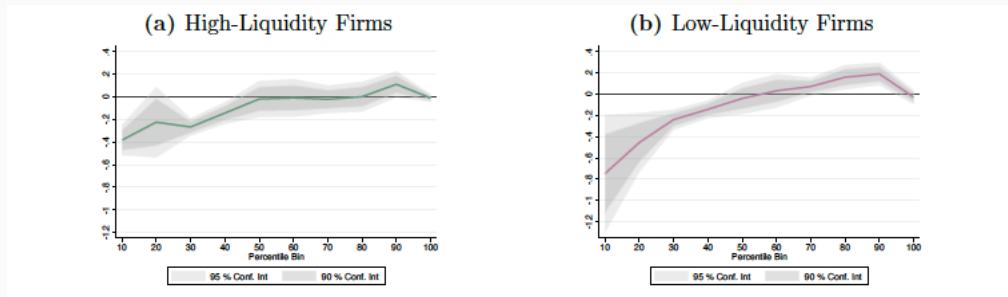


Differential Impact for Working Capital

Working capital as the ratio of short-term assets to short-term liabilities



Differential Impact for Wage Growth



The horizontal axis shows the wage deciles, and the vertical axis shows the estimated coefficient of the total effect of a positive credit-supply shock in the wage growth rate

Model

- Real Small Open Economy
- Discrete Time, Infinite Horizon
- Agents: Households (Skilled vs Unskilled), Banks (Intermediation Premium), and Firms
 - Financial Frictions: Firms face working capital constraints
- One single source of uncertainty: changes in the intermediation premium (credit supply shock)
- The labor market is divided in two separate markets, one for skilled workers and another for unskilled workers, where workers search for jobs posted by firms as in Shimer (2010)

Household

- The representative household is composed by many infinitely lived individuals of two types, skilled z and unskilled u , where each type has measure 1
- Every period, each household member $i_n \in [0, 1]$ is employed l_n or unemployed u_n , where $n \in \{z, u\}$
- The household's recursive problem is

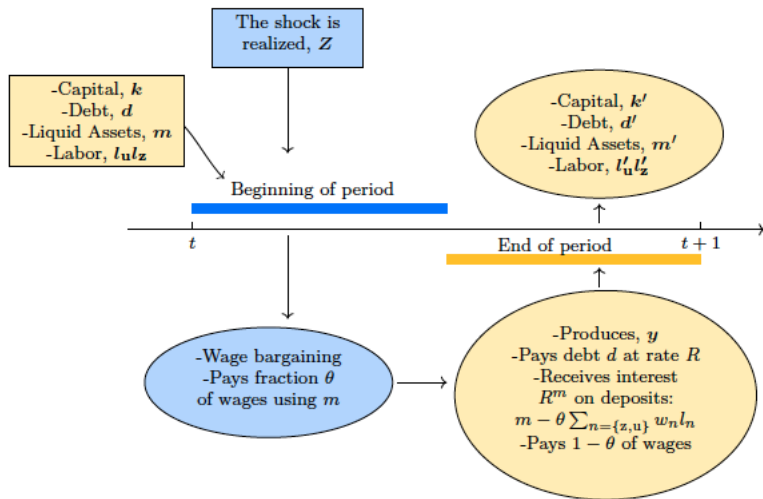
$$\begin{aligned} V^H(s, d^h, l_u, l_z) &= \max_{c, d^h} U(c, l_u, l_z) + \beta \mathbb{E} \left[V^H(s', d^{h'}, l'_u, l'_z) \right] \\ \text{s.t. } c + d^h &= w_u l_u + w_z l_z + \frac{1}{M(s'|s)} d^{h'} + \pi^F + \pi^B \\ l'_n &= (1 - \rho_n) l_n + p(\theta_n) u_n, \quad n \in \{z, u\} \end{aligned}$$

- We then have $M(s'|s) = \beta \frac{U_1(c', \cdot, \cdot)}{U_1(c, \cdot, \cdot)}$ as the sdf

- Banks are owned by households and take deposits m from firms, paying an exogenous rate $R^m > 1$ on the deposits that stay within it until the end of the term, and make loans d every period to maximize profits, s.t. intermediation costs Z
- The bank's recursive problem is

$$\begin{aligned}
 V^B(s, d, d^h, m, Z) &= \max_{d'} \pi^B + \beta M(s' | s) \mathbb{E} \left[V^B(s', d', d^{h'}, m', Z') \right] \\
 \text{s.t. } \pi^B &= Rd - d' + m' - R^m m \\
 &\quad + \theta (R^m - 1) \sum_{n=z, u} w_n l_n - Z \tau(d') \\
 \log(Z') &= \eta \log(Z) + v, \quad v \sim N(0, \sigma_Z^2)
 \end{aligned}$$

where $\tau(d)$ is the elasticity of the credit supply, assumed to be a concave function of d .



- The recursive problem of the firm is

$$\begin{aligned}
 J(s, k, d, m, l_u, l_z, Z) = & \max_{v_n, l_n, k', b', m'} \pi^F + \mathbb{E} \left[M(s' | s) J(s', k', d', m', l'_u, l'_z, Z') \right] \\
 \text{s.t. } \pi^F = & f(k, l_u, l_z) - \sum_n w_n l_n - \theta (R^m - 1) \sum_n w_n l_n - \sum_n \zeta_n v_n \\
 & - x - h(k', k) + d' - R d - \kappa(d', k) + R^m m - m' \\
 x = & k' - (1 - \delta) k \\
 m' \geq & \theta \sum_n w'_n l'_n \\
 l'_n = & (1 - \rho_n) l_n + p(\theta_n) v_n, \quad n \in \{u, z\}
 \end{aligned}$$

where

$$f(k, l_u, l_z) = \left(\mu (l_z)^{\eta_a} + (1 - \mu) (\mu_r (k)^{\eta_r} + (1 - \mu_r) (l_u)^{\eta_r})^{\frac{\eta_a}{\eta_r}} \right)^{\frac{1}{\eta_a}}$$

and $h(k', k)$ and $\kappa(d', k)$ are capital adjustment costs

- Given initial conditions k_0 , d_0 , and m_0 , contingent state s and a realization of the shock in Z_t , and a steady-state debt holdings position $\bar{d}$, an equilibrium is a sequence of allocations — k_t , c_t , d_t , m_t — and prices — $w_{zt}, w_{ut}, R, M(s'|s)$ — such that all the markets clear.
- The household holds a trade deficit with the rest of the world

- Positive credit supply shock: $Z \downarrow$
 1. From the banks' problem, $\mathbb{E} \left[M(s'|s) R' \right] = (1 + Z\tau'(d'))$. So, $M(s'|s) R' \downarrow$.
 2. From the firms' FOC for debt, $(1 - \kappa_1(d', k)) = \mathbb{E} \left[M(s'|s) R' \right]$. So, $d' \uparrow$
 3. From the firms' FOC for capital,
 $(1 + h_1(k', k)) = \mathbb{E} \left[M(s'|s) (MPK + 1 - \delta + h_2(k'', k') + \kappa_2(d'', k')) \right]$, so $k' \uparrow$
- Because firms use liquid assets to finance working capital, increasing debt demand depresses labor demand

Mechanism - Employment & Wages

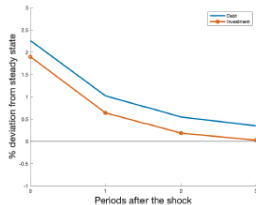
- From the firms' FOC for the liquid asset, $1 = \mathbb{E} \left[M(s'|s) (R^m + \lambda'_{f1}) \right]$, where λ_{f1} is the LM for the money-holdings constraint. So, when $Z \downarrow$, $\lambda'_{f1} \uparrow$ and $m' \downarrow$
- From the wage bargaining solution, we have

$$w_u = \left(\mu_u MPL_u + \mu_u \zeta_u \theta_u + \frac{(1-\mu)\phi l_u^{(\nu-1)}}{u_1(c, l_z, l_u)} \right) \times \frac{1}{1 + \mu_u(R^m - 1 + \lambda_{f1})\theta}$$
$$w_z = \left(\mu_z MPL_z + \mu_z \zeta_z \theta_z + \frac{(1-\mu)\phi l_z^{(\nu-1)}}{u_1(c, l_z, l_u)} \right) \times \frac{1}{1 + \mu_z(R^m - 1 + \lambda_{f1})\theta},$$

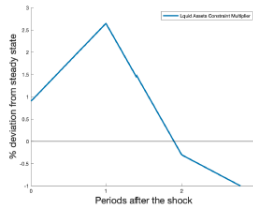
- If $\theta = 0$, wages determined solely by the PF. MPLs are both increasing on k , but average effect will depend on how much firms want to substitute low-skill workers for capital
- If $\theta > 0$, $\lambda'_{f1} \uparrow$ (i.e. $m' \downarrow$) so that labor demand decreases overall, but effect is not symmetric for all types of workers.

The Effect of Positive Credit-Supply Shock on the Firm - Model IRFs

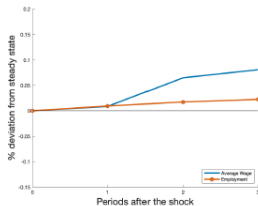
(a) Debt and Gross Investment



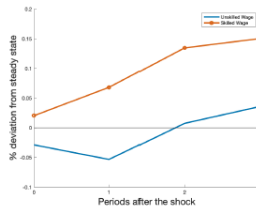
(b) Shadow Value of Liquid Assets



(c) Employment and Average Wages



(d) Low and High Wages



Conclusion and Comments

Summary of Results

1. Firms increase investment but the effect on average employment and average wages is small and insignificant
 2. Wages in the bottom half of the distribution decline → evidence of capital-low-skill substitutability?
 3. Potential Mechanism → Financial (liquidity) constraints and the degree substitutability between capital and low-skilled labor. Why?
 - Firms with highly liquid asset holdings are more responsive to the shock and expand in scale
 - Firms with illiquid asset holdings reduce their working capital in a response to a positive credit-supply shock and if anything, reduce their labor demand
- Model with simple financial frictions

Comments

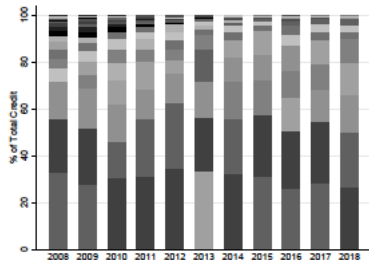
- Interesting idea, but execution is not there yet
- At the moment, the paper relies on its empirical analysis, but some issues
 1. A deeper discussion of the identifying assumptions. Is there any validation possible with price (interest rate) data?
 2. It does not take the role of financial heterogeneity seriously enough when it could do so
 - With this level of detail, why not measure the interaction between financial stance of the firm and the response to the credit supply shock directly as in Ottonello & Winberry (2020)?
 - Unpack the impact on the working capital of firms with lower liquidity: is it being driven mostly by movements in the asset structure or in the liability structure?
- The model does not seem to be playing a relevant role in the paper
 - If financial heterogeneity matters, why does not the model takes this into account seriously?
 - How does the model fare with regards to the data? Model fit comparison against

Appendix

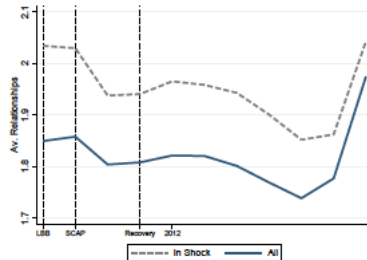
1. The link between financial shocks and labor markets
 - Berton et al. (2018); Huber (2018); Popov and Rocholl (2018); Chodorow-Reich (2014)
2. The link between financial shocks and firm dynamics
 - Amiti and Weinstein (2018); Jimenez et al. (2019); Gilchrist et al. (2017); Khwaja and Mian (2008); Kim (2018)
3. The importance of financial frictions in small open economies
 - Neumeyer and Perri (2005); Quadrini (2011); Leyva and Urrutia (2020)

Identifying Assumptions of the Supply Shock – Credit Market Structure

(a) Credit Concentration



(b) Number of Banking Relationships



Cross-Validation of the Banking Shock – Correlated with Banks' Characteristics

	(1)	(2)	(3)	(4)
	$\hat{\beta}_{bt}$			
$\Delta \log$ Comm. credit	0.27*** (0.08)			
Dividends dummy		0.34** (0.06)		
CASA			0.36*** (0.14)	
Capital to liabilities				-0.43 *** (0.22)
Time FE	Yes	Yes	Yes	Yes
<i>N</i>	145	145	145	145

Note: Each column estimates $\hat{\beta}_{bt} = \eta_1 + \eta_2 y_{bt} + \alpha_t + \epsilon_{bt}$, where y_{bt} is a bank-level outcome (Change in credit, CASA ratio, Dividends dummy, Capital adequacy), and α_t are time fixed effects. Robust standard errors in parentheses clustered at the bank level.* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Cross-Validation of the Banking Shock – Uncorrelated with the Business Cycle

	(1)	(2)	(3)	(4)
	$\hat{\beta}_{bt}$			
$\hat{\beta}_{bt-1}$	0.36*** (0.12)	0.36*** (0.12)	0.36*** (0.12)	0.37*** (0.12)
Cyclical component GDP		0.39 (0.87)		
Cyclical component GDP _{t-1}			-0.02 (0.91)	
Cyclical component GDP _{t+1}				1.24 (0.93)
Cons	0.00 (0.02)	-0.01 (0.01)	0.00 (0.02)	-0.02 (0.02)
N	137	137	137	123

Note: Each column estimates $\hat{\beta}_{bt} = \eta_1 + \rho\hat{\beta}_{bt-1} + \eta_2 y_{bt} + \epsilon_{bt}$, where y_{bt} is the cyclical component of GDP using the HP filter with smoothing parameter $\lambda = 400$. Column 2 uses on impact GDP, Column 3 uses lagged GDP, and Column 4 uses forward GDP. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Distributional Impact of Credit Shocks

- Let $p(w_{ifj})$ denote each decile of the distribution of wages
- Then, we estimate

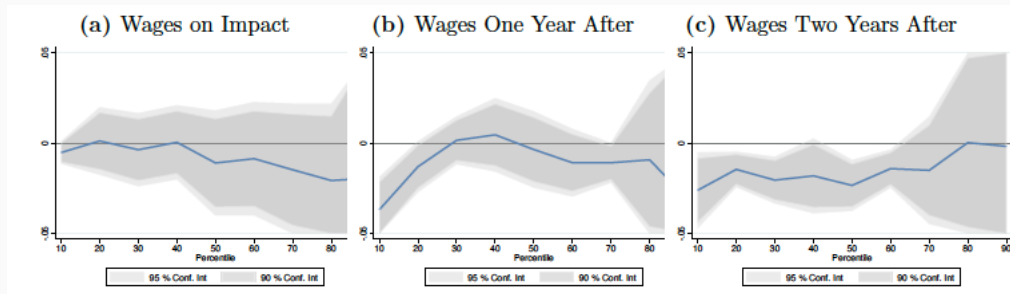
$$p(\log(w_{ift+h})) = \beta_0 + \beta_s \text{Supply Shock}_{ft} + X_{ift-1}\Gamma + \alpha_{fth} + \alpha_i + \varepsilon_{ifth}$$

where β_s is the coefficient of interest

- To keep the analysis comparable with the Jordà projections at the firm level, they study the effect of a shock in t on the distribution of wages one, two, and three years after the shock

Distributional Impact of Credit Shocks - Results

A positive credit-supply shock reduces the value of the wage deciles on the bottom half of the distribution



The horizontal axis shows the wage deciles, and the vertical axis shows the change in each decile's cutoff value